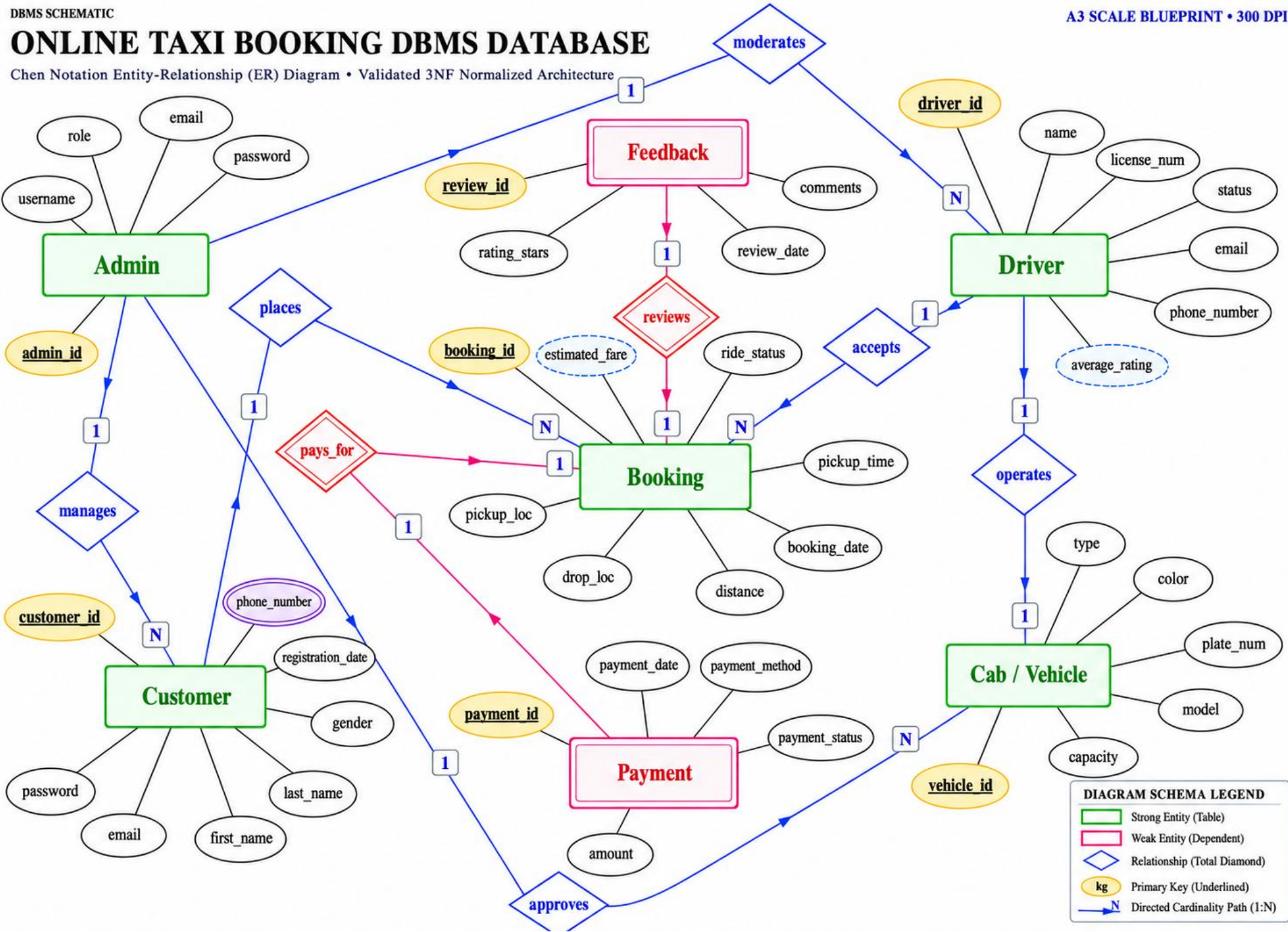
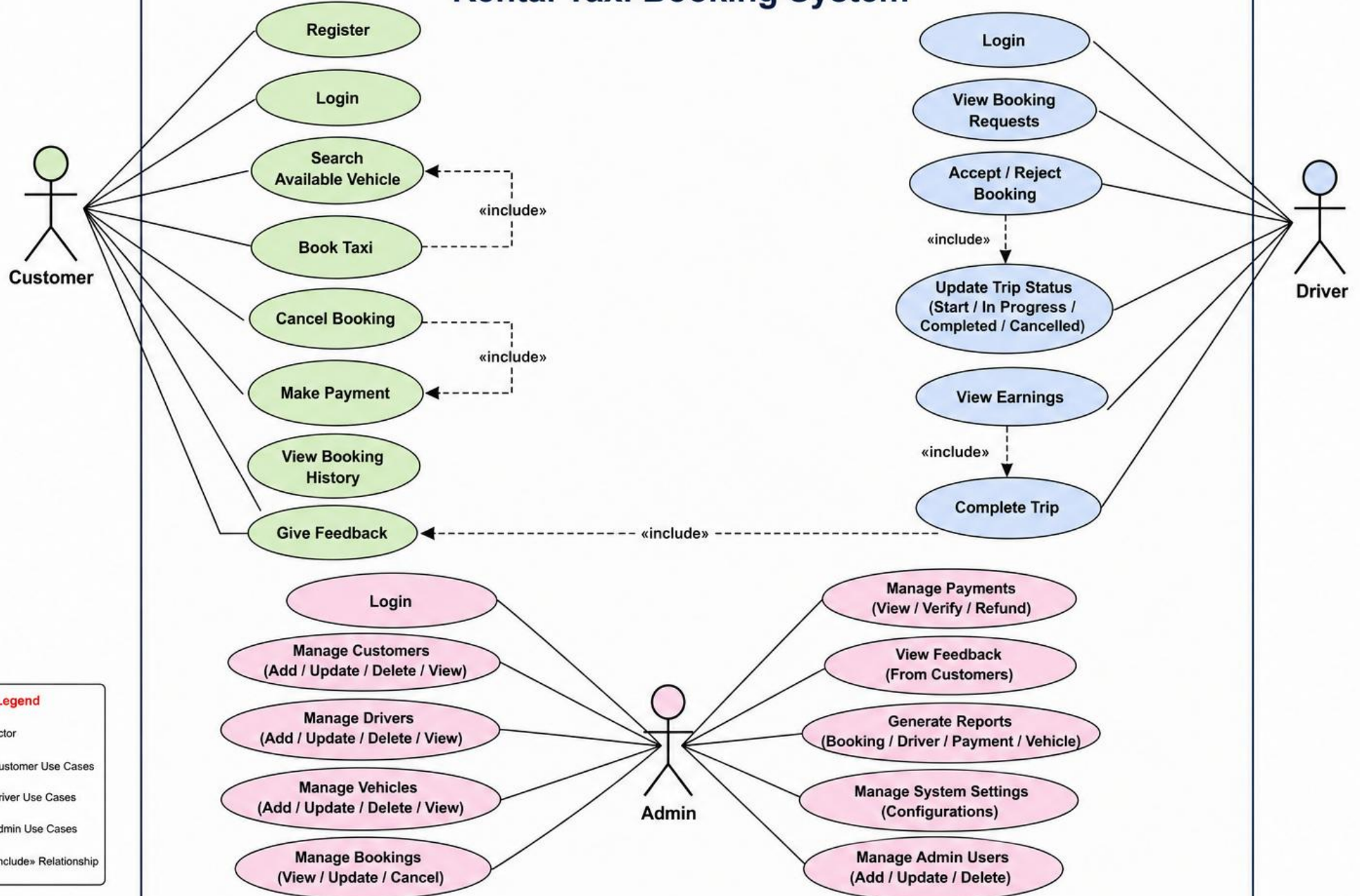


ONLINE TAXI BOOKING DBMS DATABASE

Chen Notation Entity-Relationship (ER) Diagram • Validated 3NF Normalized Architecture

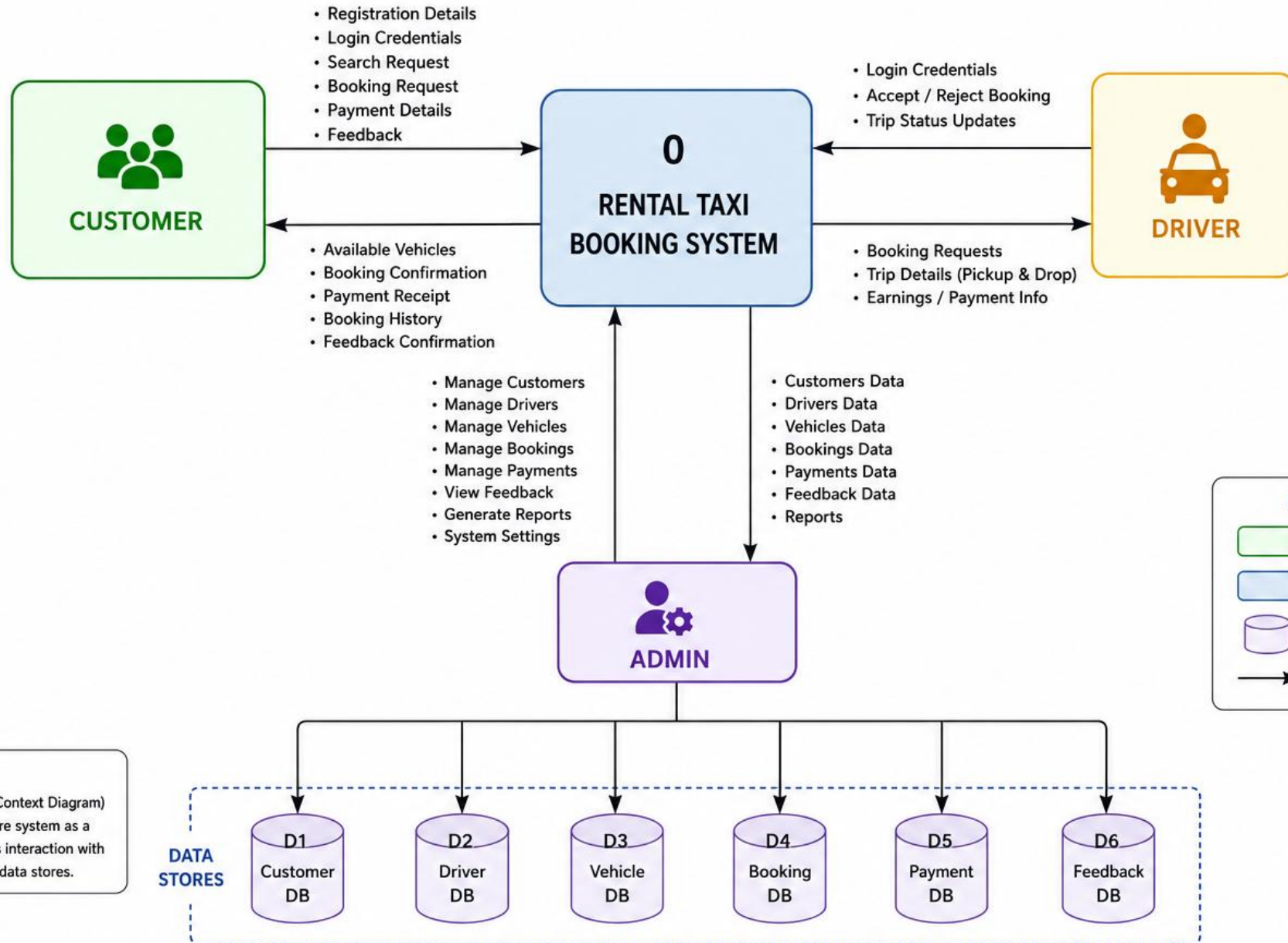


Rental Taxi Booking System



LEVEL 0 DATA FLOW DIAGRAM (CONTEXT DIAGRAM)

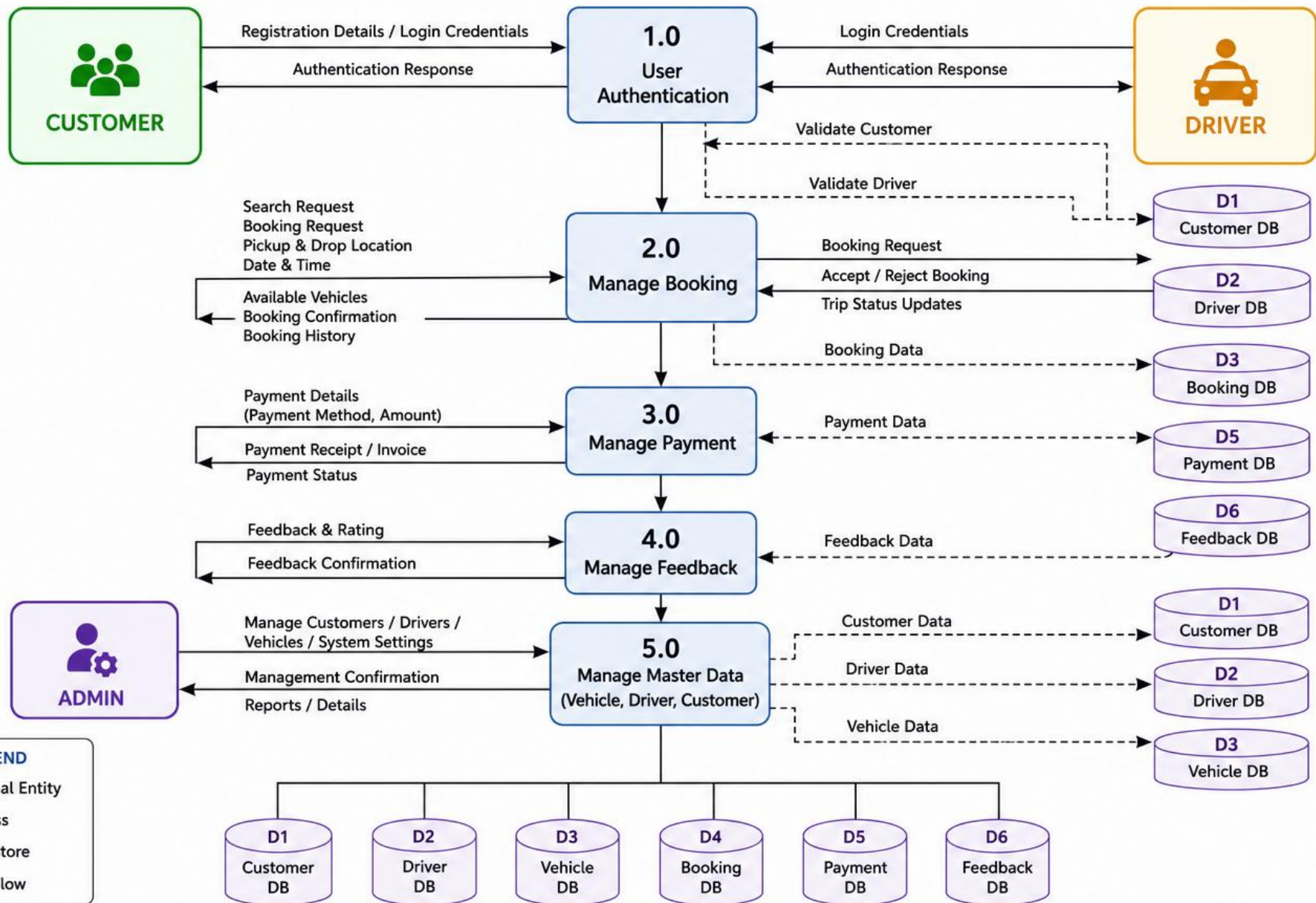
RENTAL TAXI BOOKING SYSTEM



NOTES:

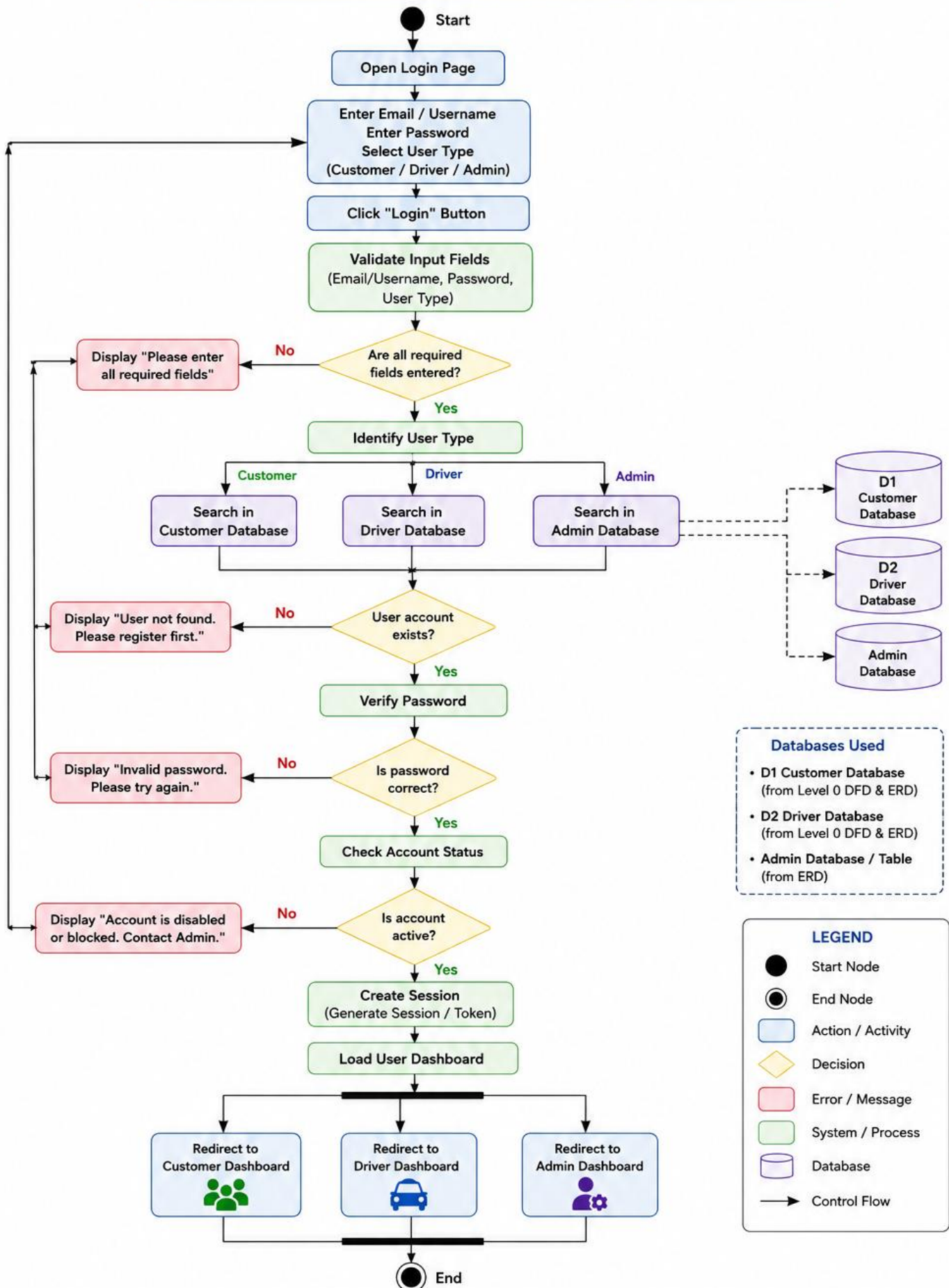
This is Level 0 DFD (Context Diagram) which shows the entire system as a single process and its interaction with external entities and data stores.

LEVEL 1 DATA FLOW DIAGRAM RENTAL TAXI BOOKING SYSTEM



LOGIN ACTIVITY DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This login process is common for all three actors: Customer, Driver and Admin.

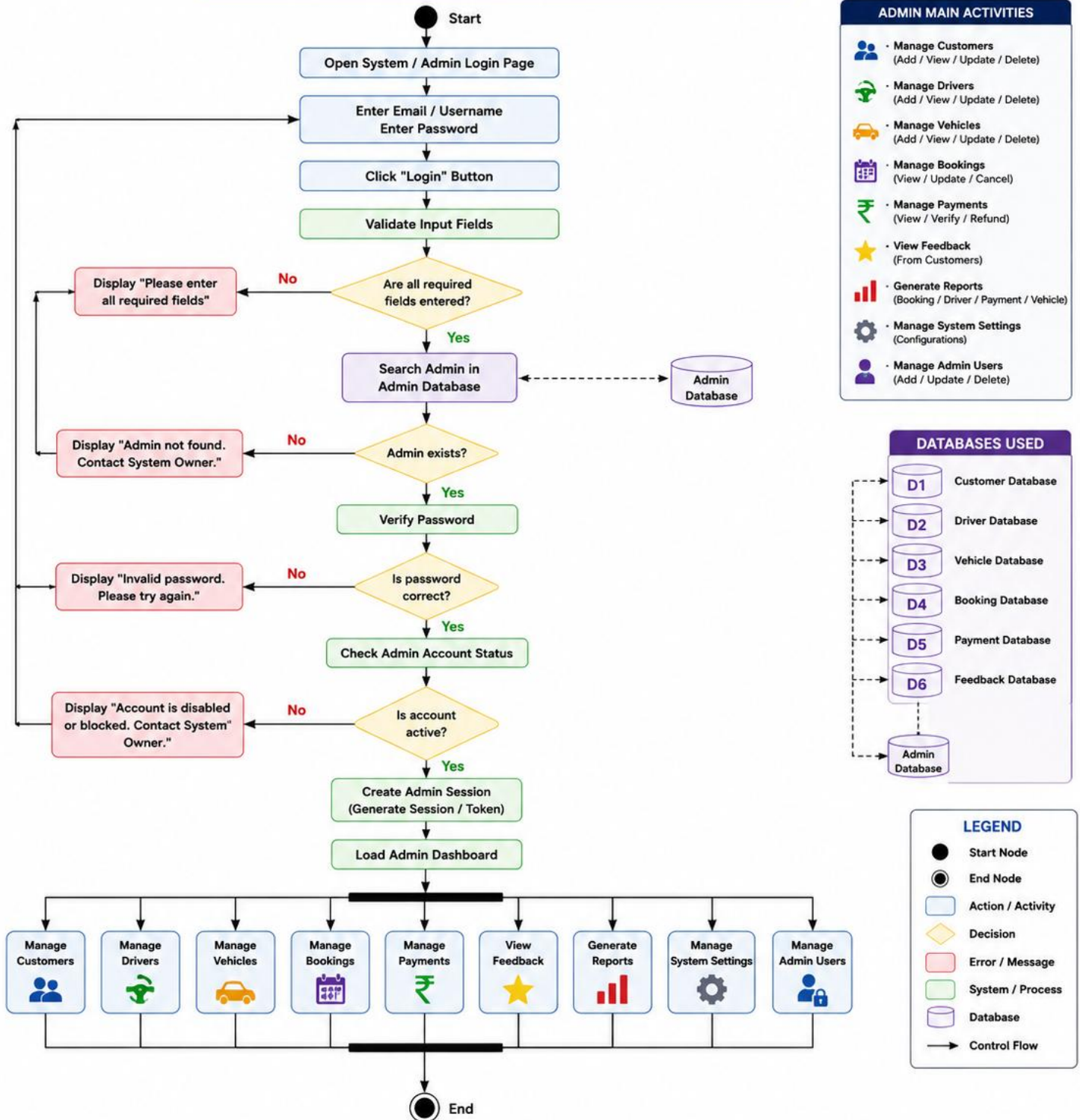


NOTES

- This activity diagram is based on Use Case Diagram (Login Use Case), Level 0 DFD (Process 1.0 User Authentication) and Level 1 DFD (Validate Customer / Driver, Authentication).
- All three actors (Customer, Driver, Admin) follow the same login process with role based redirection to dashboard.

ADMIN ACTIVITY DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This diagram shows the activities performed by Admin in the system.

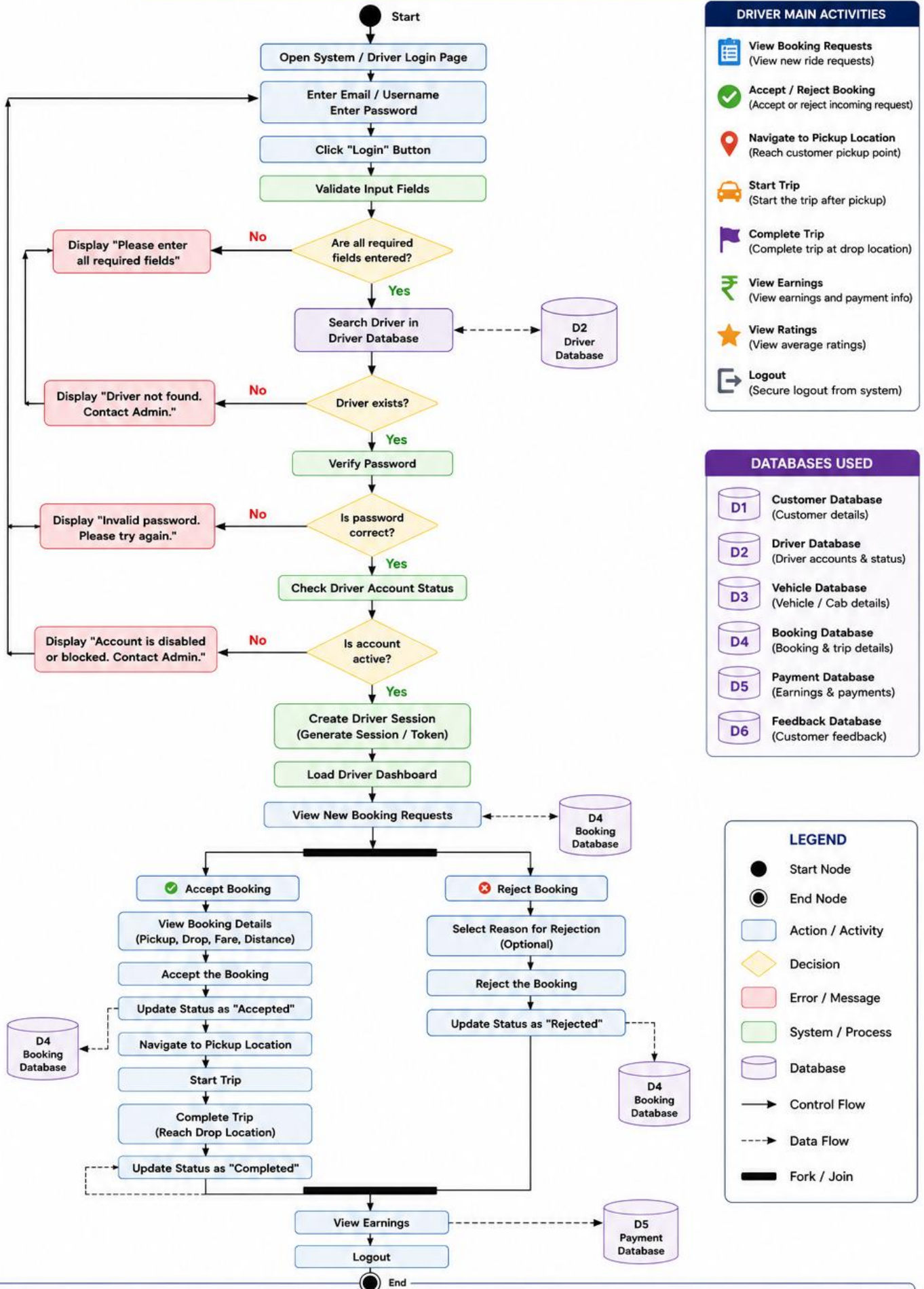


NOTES:

- This activity diagram is based on Use Case Diagram (Admin use cases), Level 0 DFD (Admin interactions) and Level 1 DFD (Process 5.0 Manage Master Data and related processes).
- Admin can perform full system operations including master data management, booking management, payment management, feedback, reports and system configuration.

DRIVER ACTIVITY DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This diagram shows the activities performed by Driver in the system.

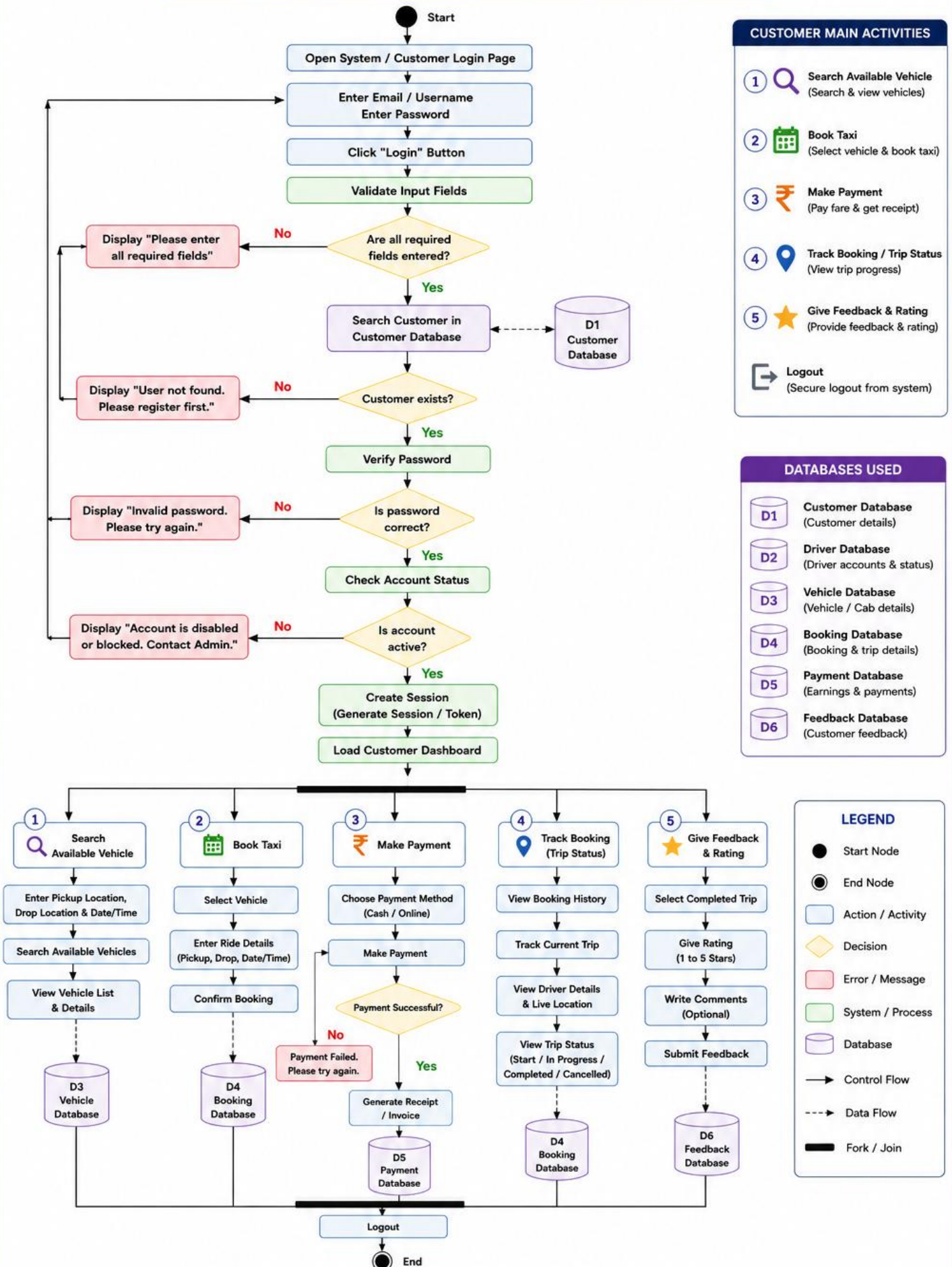


NOTES

- This activity diagram is based on Use Case Diagram (Driver use cases), Level 0 DFD (Driver interactions) and Level 1 DFD.
- Driver can accept or reject booking requests. If accepted, driver completes the trip and earnings are recorded.
- All updates (accepted, rejected, completed) are stored in Booking Database and earnings in Payment Database.

CUSTOMER ACTIVITY DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This diagram shows the activities performed by Customer in the system.

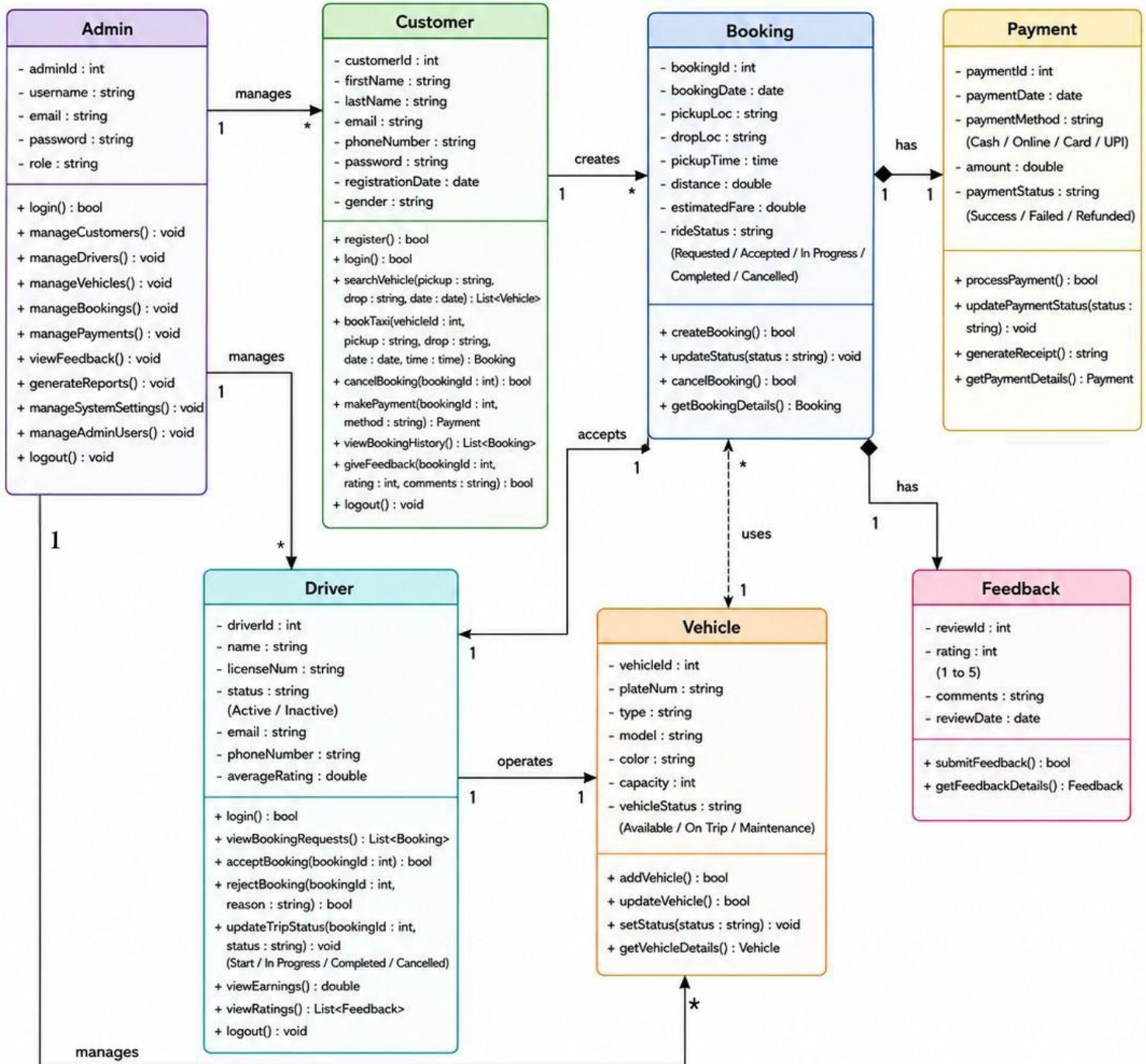


NOTES:

- This activity diagram is based on Use Case Diagram (Customer use cases), Level 0 DFD (Customer interactions) and Level 1 DFD (Processes 1.0, 2.0, 3.0, 4.0 used by customer).
- Customer can search & book taxi, make payment, track trips and give feedback.
- All data is stored in respective databases as shown in the diagram.

CLASS DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This diagram represents the classes in the system, their attributes, operations (methods) and the relationships between them.



LEGEND

- Association
- - - - Dependency (Uses)
- ◆ Composition
- 1 One
- * Many

DATABASES (From DFD & ER Diagram)

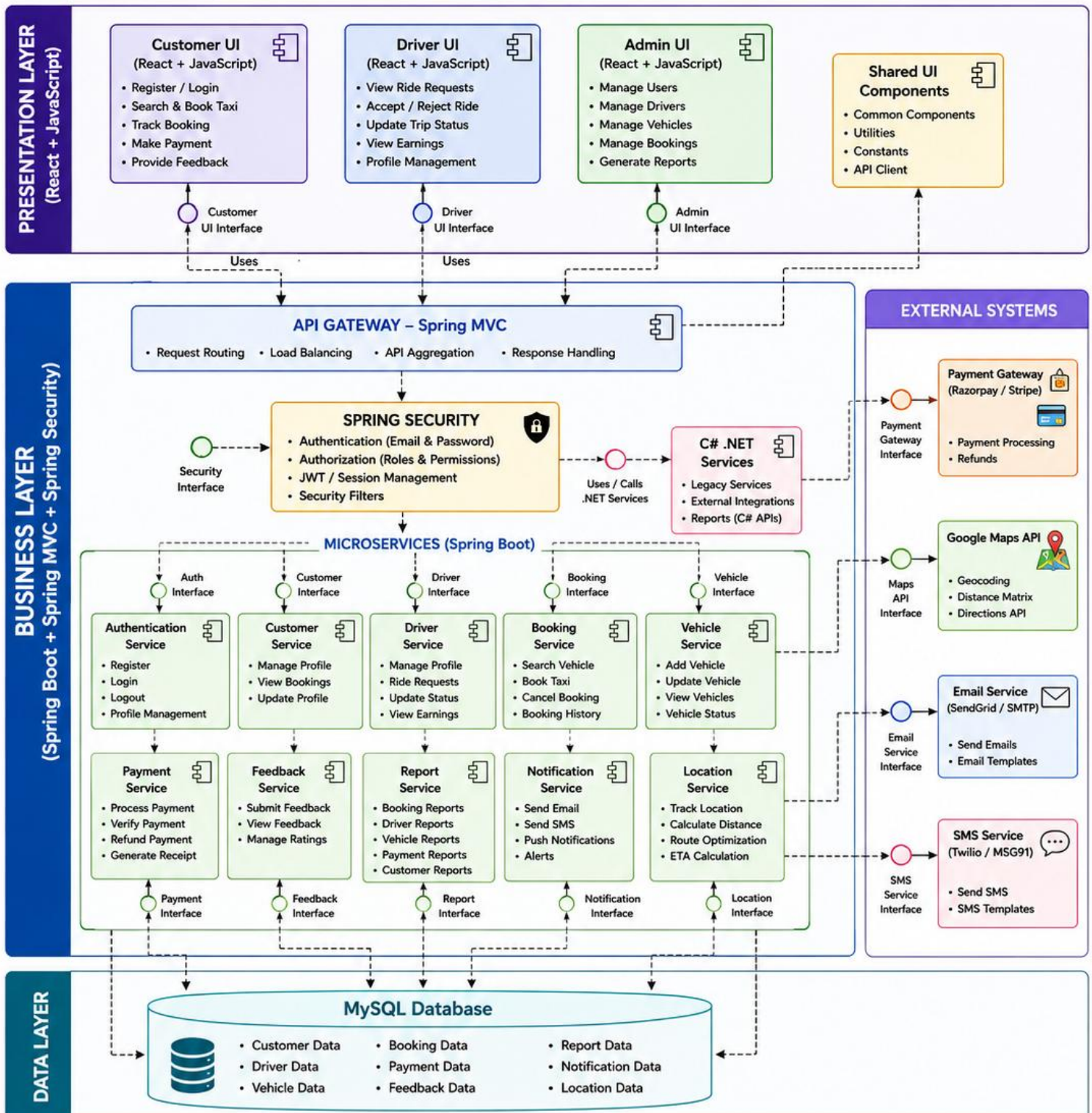
- D1 Customer Database
- D2 Driver Database
- D3 Vehicle Database
- D4 Booking Database
- D5 Payment Database
- D6 Feedback Database

NOTES

- This class diagram is derived from ER Diagram, Use Case Diagram and DFDs (Level 0 & Level 1).
- Admin manages all master data, bookings, payments, reports and system settings.
- Customer can search vehicle, book taxi, make payment, track bookings and give feedback.
- Driver can view booking requests, accept/reject, update trip status and view earnings & ratings.
- Booking is created by Customer, accepted by Driver, uses Vehicle and has Payment and Feedback (1:1 with both).

COMPONENT DIAGRAM – RENTAL TAXI BOOKING SYSTEM

This diagram shows the major software components, their provided/required interfaces and the dependencies between them.



LEGEND

- Component
- Provided Interface
- Required Interface
- Assembly / Communication
- Dependency (Uses)

NOTES

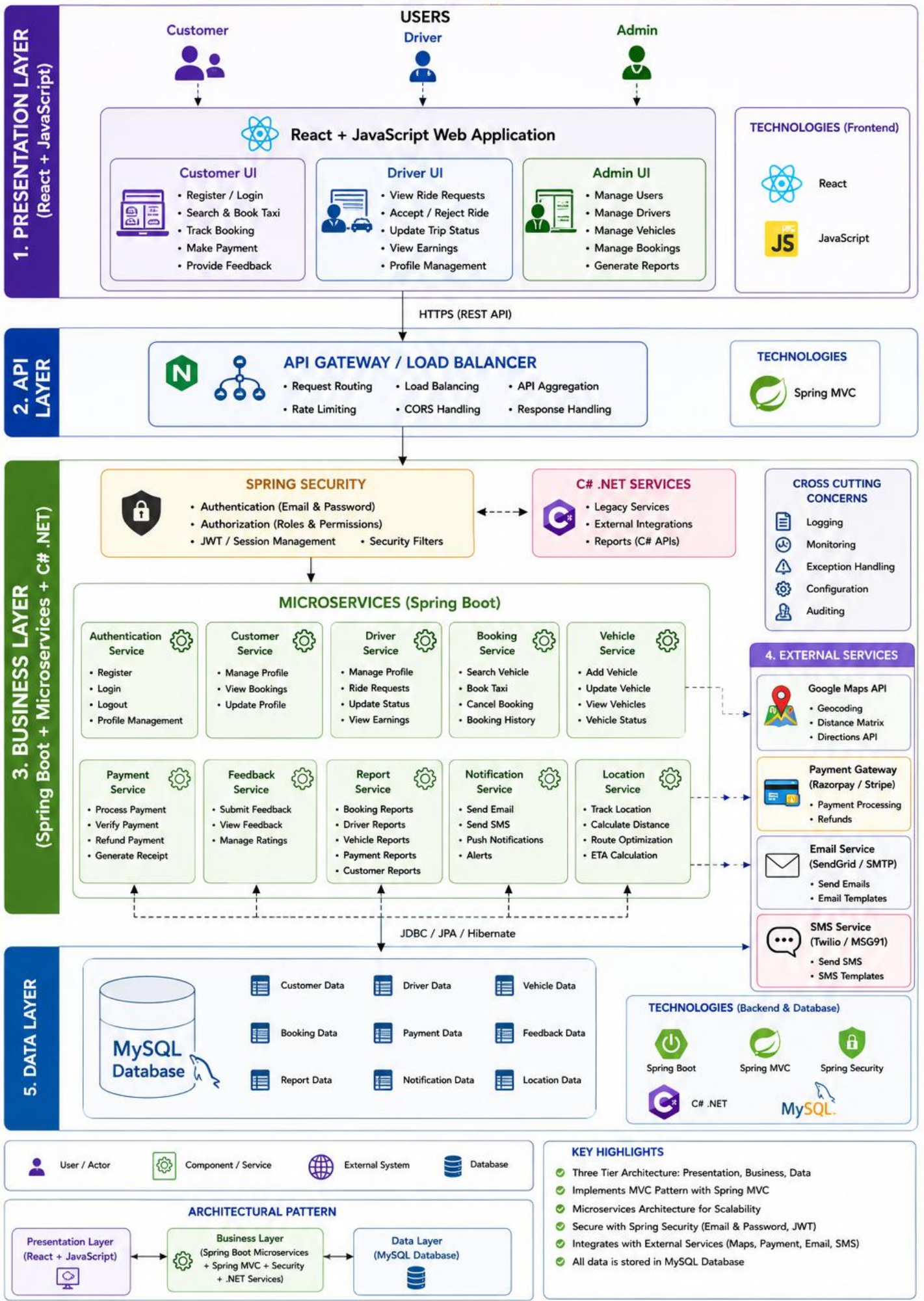
- The system follows Three-Tier Architecture (Presentation, Business, Data).
- Presentation Layer is built with React + JavaScript.
- Business Layer uses Spring Boot, Spring MVC, Spring Security and Microservices.
- Some legacy/reporting services are implemented using C# .NET.
- All services communicate through REST APIs.
- External services (Payment Gateway, Maps, Email, SMS) are integrated via APIs.
- All persistent data is stored in MySQL Database.

TECHNOLOGIES



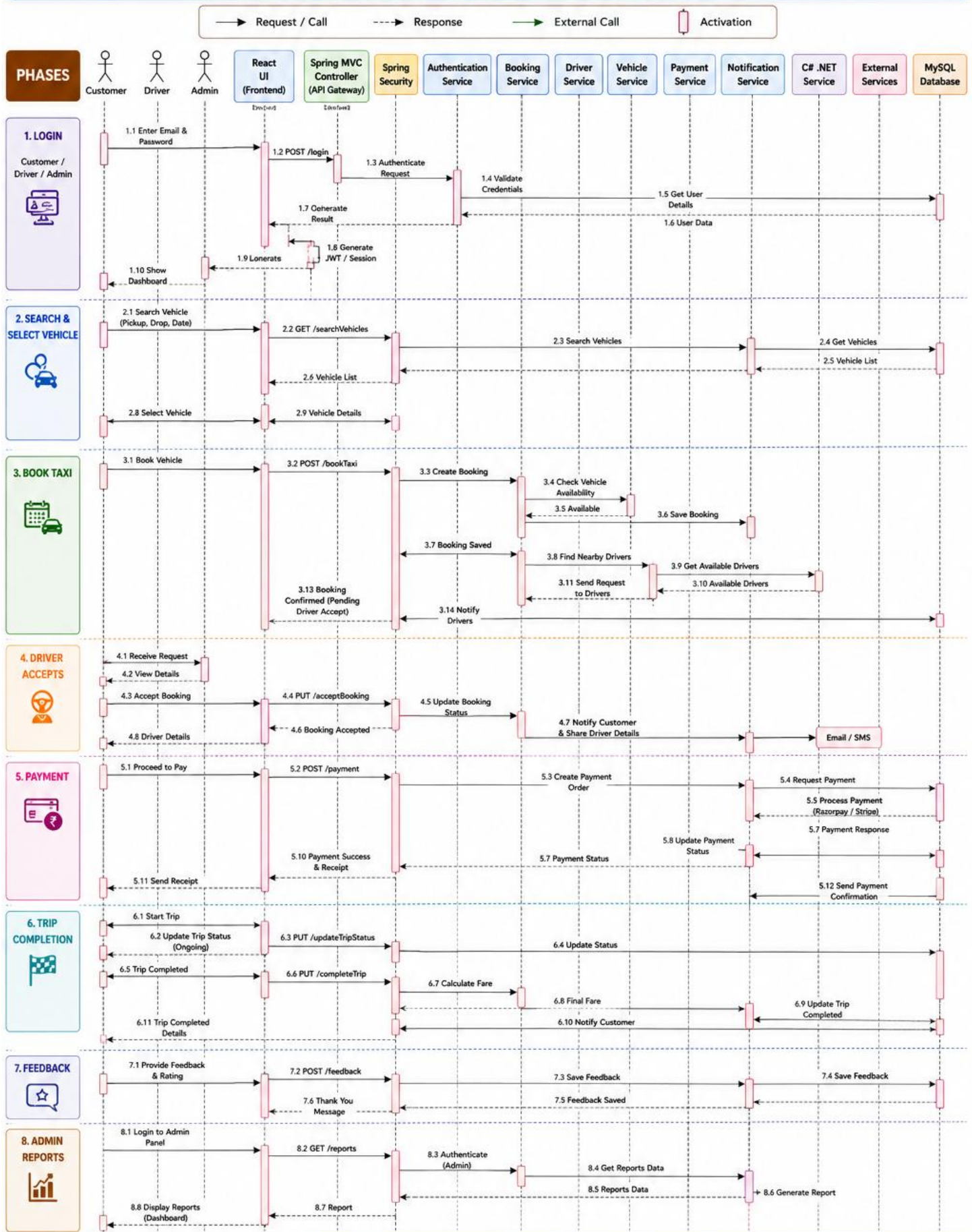
ARCHITECTURE DIAGRAM – RENTAL TAXI BOOKING SYSTEM

High Level Architecture showing the structure, layers, services, databases and external systems of the Rental Taxi Booking System.



SEQUENCE DIAGRAM – RENTAL TAXI BOOKING SYSTEM

(All Major Use Cases Clubbed in One Sequence Diagram)



LEGEND

- Actor
- System / Component
- Activation Bar
- LifeLine

NOTES

- This sequence diagram shows the complete flow of the system from Login to Reports.
- All main use cases are clubbed in a single diagram.
- Solid arrows represent requests/calls and dashed arrows represent responses.
- External services include Payment Gateway, Email Service, SMS Service and Google Maps API.

EXTERNAL SERVICES

- Google Maps API (Maps, Geocoding, Distance)
- Payment Gateway (Razorpay / Stripe)
- Email Service (SendGrid / SMTP)
- SMS Service (Twilio / MSG91)

TECHNOLOGIES USED

- React
- JavaScript
- Spring Boot
- Spring Security
- Spring MVC
- C# .NET
- MySQL

ONLINE TAXI BOOKING DBMS DATABASE DESIGN DIAGRAM

A3 SCALE DESIGN • 300 DPI

